

Sharif University of Technology
Department of Mechanical Engineering

Modeling of an RRP Robot in the Simulink Environment

Robotics Course – Homework 3

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1 Problem Statement

We are first asked to simulate the RRP robot model in the MATLAB Simulink environment and to show the reference coordinate frames on the given figure.

1.1 Question 1

In the first part of the problem, given values must be applied as initial conditions to each of the joints, and then, for each case, the coordinates of the end effector must be found with respect to the base (zero) frame.

1.2 Question 2

In the second part, a function must be defined in MATLAB that is able to solve the inverse kinematics of this robot. The input of this function is the coordinates of the end effector in the base frame, and its output is the three joint variables.

2 Answers

2.1 Question 1

First, the links were drawn in SolidWorks, and then they were opened in the Simulink environment using the **Solid File** block. All parts are taken to be made of steel, and its density is entered manually in MATLAB:

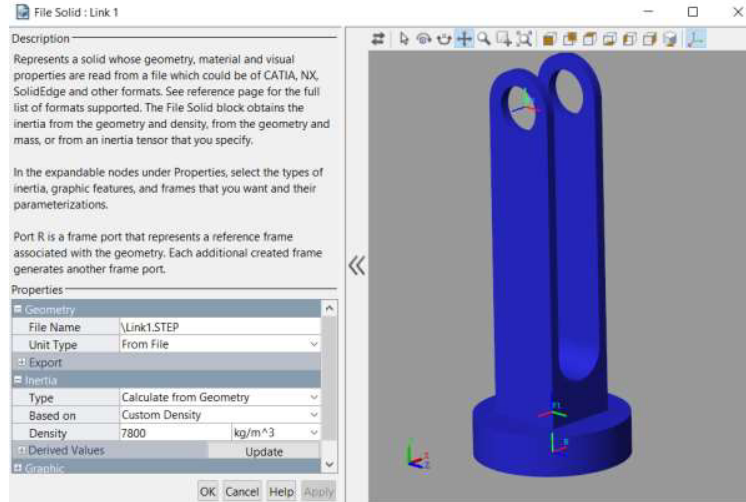


Figure 1: File Solid block settings for Link 1 (imported from `Link1.STEP`, density 7800 kg/m^3).

Reference frames appropriate to the use of each link were mounted on it. Then, using two Revolute joints and one Prismatic joint, the given mechanism was assembled.

We use the **Targets: State** and **Target: Specify Position** sections in the settings of each joint, and in the initial state we set the conditions as below, so as to have an overall view of the mechanism. The figure below shows the mechanism in the state where all three variables θ_1 , θ_2 , and d are equal to zero.

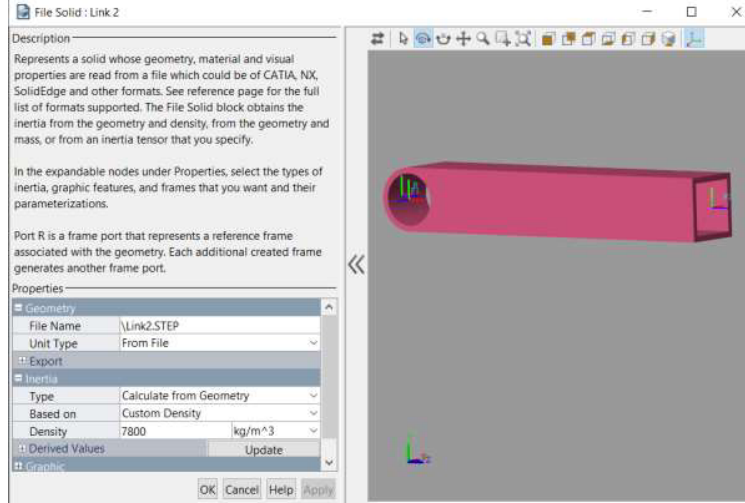


Figure 2: File Solid block settings for Link 2 (imported from Link2.STEP).

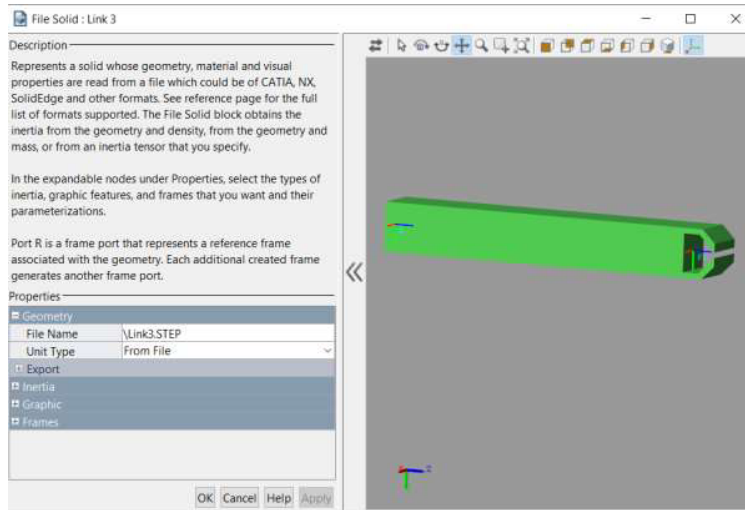


Figure 3: File Solid block settings for Link 3 (imported from Link3.STEP).

Now, for the three cases given below, using **Targets: State** in the settings of each joint, the given positions are set, and the position of the end effector with respect to the base frame is recorded for each case. To read the position of the end effector, we use the **Transform Sensor** block:

Mode	1	2	3
θ_1	130	30	90
θ_2	10	50	20
d	20	80	250
x (mm)	-241.4	-122.1	-516.8
y (mm)	-202.6	211.5	0
z (mm)	365.6	601.1	498.1

We check the correctness of these computations by carrying out the Denavit–Hartenberg procedure on the given mechanism. The D-H table is formed as follows:

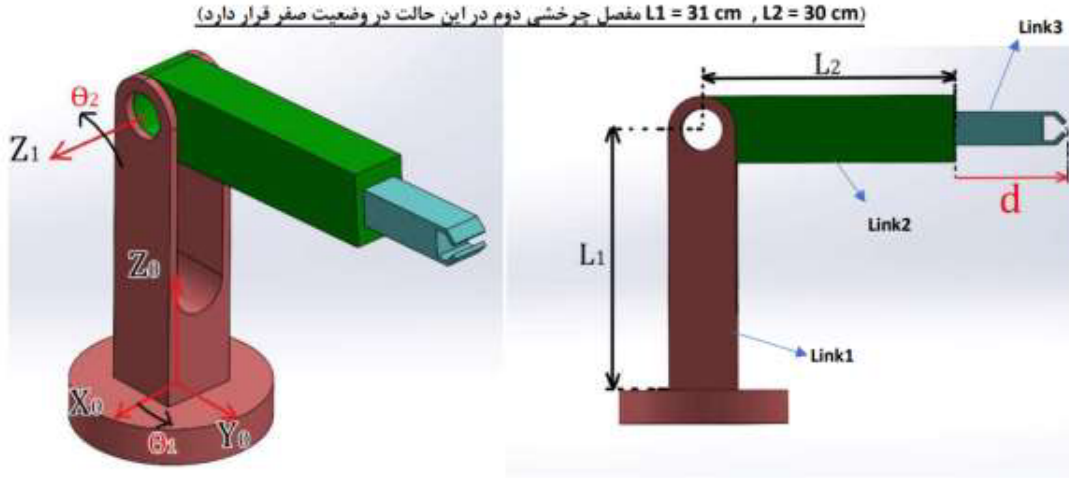


Figure 4: Link geometry and reference frames of the RRP robot, drawn in SolidWorks ($L_1 = 31 \text{ cm}$, $L_2 = 30 \text{ cm}$; the second revolute joint is at zero position in this state).

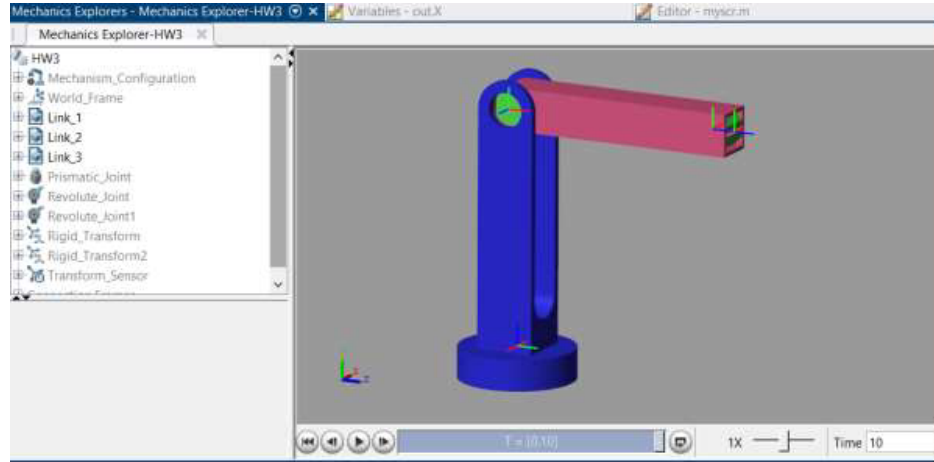


Figure 5: Simulink Mechanics Explorer view of the assembled mechanism with $\theta_1 = \theta_2 = d = 0$.

	θ	d	l	α
1	θ_1	310	0	0
2	90	0	0	90
3	θ_2	0	$300 + d$	0

We write the following MATLAB code to compute the coordinates of the end effector for the three given cases:

```
theta1 = [130,30,90];
theta2 = [10,50,20];
d = [20,80,250];

P30 = ones(4,3);
H03 = ones(4,4);

for i = 1:3
```

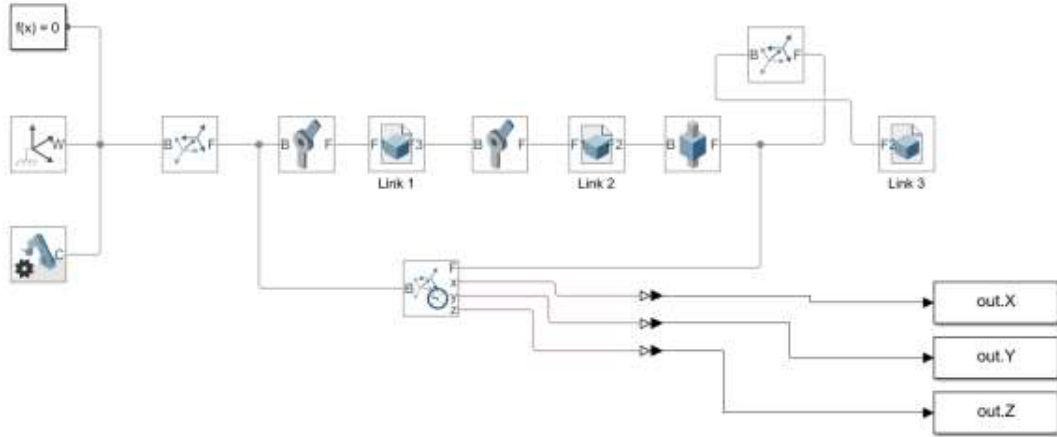


Figure 6: Simulink block diagram of the assembled RRP mechanism, showing the Link 1–3 File Solid blocks, the joints, the Transform Sensor, and the outputs out.X, out.Y, out.Z.

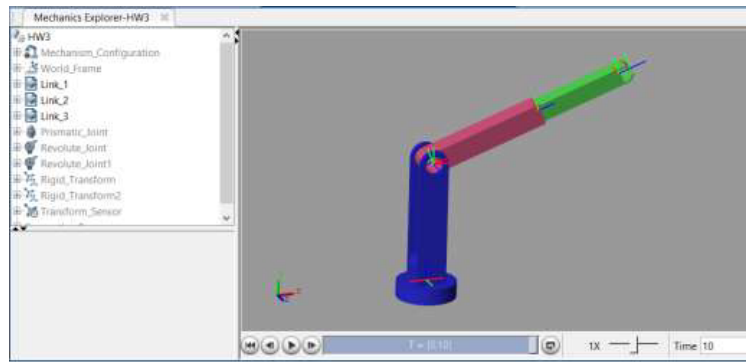


Figure 7: Mechanics Explorer view of the mechanism in Mode 3.

```

H30 = Rot('Z',theta1(i))*Trans('Z',310)*Rot('Z',90)*Rot('X',90)*Rot('Z',
    'theta2(i))*Trans('X',300+d(i));
P33 = [0;0;0;1];
P30(:,i) = H30*P33;
end
P30

```

The final output of the code is as follows:

```

P30 =
-241.4101  -122.1296  -516.8309
-202.5671   211.5348         0
 365.5674   601.0969   498.1111
   1.0000    1.0000    1.0000

```

It can be seen that our result matches the values given by Simulink.

2.2 Question 2

We use the MATLAB Symbolic Toolbox to solve this problem. To this end, we first obtain the parametric solution of the robot's forward kinematics using the following code.

```
syms theta1;
syms theta2;
syms d;

H30 = Rot('Z',theta1)*Trans('Z',310)*Rot('Z',90)*Rot('X',90)*Rot('Z',
    theta2)*Trans('X',300+d);

H30*[0;0;0;1]
```

The answer given by MATLAB is as follows:

```
ans =
-cos((pi*theta2)/180)*sin((pi*theta1)/180)*(d + 300)
 cos((pi*theta1)/180)*cos((pi*theta2)/180)*(d + 300)
      sin((pi*theta2)/180)*(d + 300) + 310
                                     1
```

The reason the trigonometric functions appear in the above form is the use of the `sind` and `cosd` functions in the definitions of the `Rot` and `Trans` functions.

We solve the robot's inverse kinematics with the function below. For each variable, two families of solutions are obtained, and we have:

```
function [theta1_1,theta1_2,theta2_1, theta2_2 ,d1, d2] = InvKin(x,y,z)

theta1_1 = (180/pi)*atan(x/y);
theta1_2 = 180-theta1_1;
%c = d + 300 , b = cos theta 2
alpha = sind(theta1_1);
beta = cosd(theta1_1);

c = sqrt(((x^2-y^2)/(alpha^2-beta^2))+(z-310)^2);

d1 = c - 300;
d2 = -c - 300;

b = sqrt((x^2-y^2)/(alpha^2-beta^2)/c^2);
b1 = b; b2 = -b;

theta2_1 = (180/pi)*acos(b1);
theta2_2 = (180/pi)*acos(b2);

end
```

We test the correctness of the function above by feeding it the end-effector coordinates from the first part:

```
[theta11,theta12,theta21,theta22,d1,d2] = InvKin
(-241.4101,-202.5671,365.5674)
```

The answer returned by the code is as follows:

```

theta11 =
    50.0000
theta12 =
   130.0000
theta21 =
    10.0000
theta22 =
   170.0000
d1 =
    20.0000
d2 =
   -620.0000

```

It can be seen that θ_{12} , θ_{21} , and d_1 are the same variables given in the statement of Question 1. We repeat this procedure for the other two cases as well. The results are given in the table below:

Mode	1	2	3
x (mm)	-241.4	-122.1	-516.8
y (mm)	-202.6	211.5	0
z (mm)	365.6	601.1	498.1
θ_{11}	50	-29.9981	-90
θ_{12}	130	209.9981	270
θ_{21}	10	50.0055	20
θ_{22}	170	129.9945	160
d_1	20	79.9735	249.9671
d_2	-620	-679.9735	-849.9671

Now we fill in the table given by Question 2 using the code above and write all the answers:

Mode	1	2	3
x (mm)	68.4	298.5	184.9
y (mm)	12.0	-250.5	139.3
z (mm)	703.9	535	86.3
θ_{11}	80.0494	-49.9968	53.0064
θ_{12}	99.9506	229.9968	126.9936
θ_{21}	80.0015	30.0018	44.0182
θ_{22}	99.9985	149.9982	135.9818
d_1	99.9747	149.9750	21.9226
d_2	-699.9747	-749.9750	-621.9226

The code and related files are included in the appendix (see the `code` directory).